While regression examined the relationship between listing price, distance and other characteristics, clustering was used to identify distinct groups of Airbnb listings with similar location and property characteristics, providing a complementary perspective on the patterns within the dataset. **[REF: clustering/unsupervised learning]**.

The optimal number of clusters was evaluated using the elbow method **[REF]** and average silhouette scores **[REF]**. Although k = 2 produced the highest silhouette score (0.392), this solution was considered too coarse to provide meaningful segmentation beyond the existing price–distance findings. The three-cluster solution achieved the next highest substantively meaningful silhouette score (0.252) and corresponded with the point at which improvements in within-cluster variation began to flatten; therefore, k = 3 was selected.

K-means clustering **[REF]** was performed on six scaled variables **[REF: standardisation/feature scaling]**. Figure X uses principal component analysis (PCA) **[REF: principal component analysis (PCA)]** to visualise the resulting clusters in two dimensions; clustering was not performed on the principal components themselves. Together, PC1 and PC2 explain 52.5% of the variance. The clusters occupy distinguishable regions of the PCA space, although some overlap remains.

Figure X: PCA projection of K-means Airbnb listing clusters (k = 3)



(Source: Team F analysis of AB_NYC_2019 dataset using R.)

Cluster profiling (Table 1) indicates that Cluster 1 represents a smaller professional/long-stay segment, characterised by a 30-night median minimum stay, high availability (316 days) and hosts with a median of 16 listings. Cluster 2 represents active value short-stay listings, with the lowest median price ($95), greatest distance from Times Square (8.7 km) and highest review activity (2.12 reviews/month). Cluster 3 comprises the largest segment (54.6%), characterised by zero median availability and low review activity.

Table 1: Profile Characteristics of the Three K-Means Airbnb Listing Clusters

	Cluster 1	**Cluster 2**	**Cluster 3**
Size	4,594 (9.4%)	17,608 (36.0%)	26,682 (54.6%)
Median price	$150	$95	$115
Median distance from Times Sq.	3.9 km (closest)	8.7 km (furthest)	5.7 km (middle)
Median minimum nights	**30**	2	3
Median # reviews	1	28	2
Median reviews/month	0.09	2.12	0.12
Median availability (days/365)	**316**	197	**0**
Median host listings count	**16**	1	1
Dominant room type	Entire home/apt (68%)	Private room (54%)	Entire home/apt (55%)
Dominant borough	Manhattan (70%)	Brooklyn (44%), spread across boroughs	Manhattan (49%) / Brooklyn (43%)

(Source: Team F analysis of the AB_NYC_2019 dataset using R.)

Overall, cluster prices decline with increasing distance from Times Square ($150, $115 and $95), supporting the broad regression pattern while demonstrating that listing segments also differ substantially in host, availability and booking characteristics. However, the moderate silhouette score and overlap in the PCA projection indicate that these segments should be interpreted as broad listing profiles rather than sharply separated groups.